

Aristotle's Organon

From Dialectical Practice to Systematic Formal Logic

History of Logic

Primary source: William and Martha Kneale, *The Development of Logic*, Chapter II, "Aristotle's Organon."

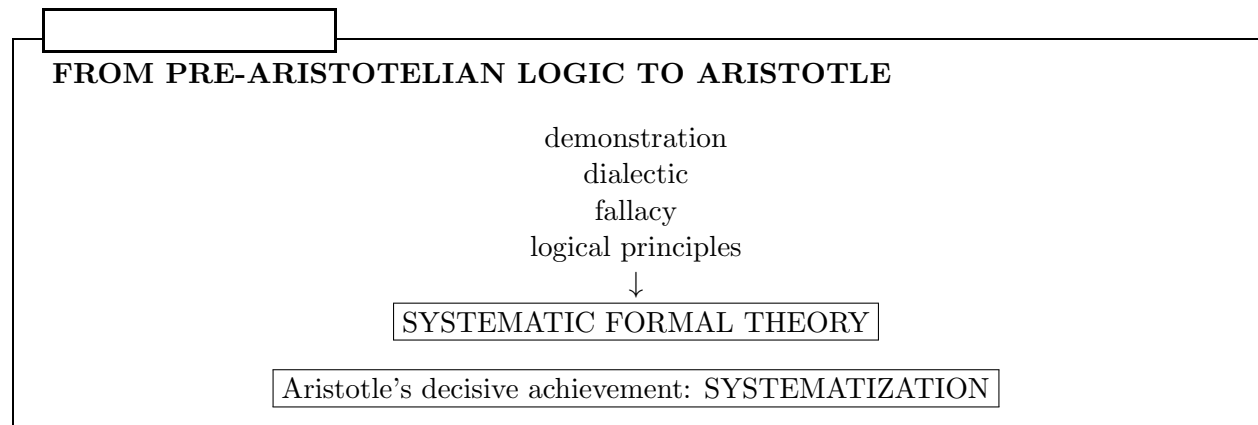
1 Aristotle and the Systematization of Logic

In the preceding lecture, we saw that substantial reflection on valid inference existed before Aristotle.

Greek geometers had developed the ideal of demonstration. Zeno and Socrates had used refutation through consequences. Sophists and Megarians had raised problems concerning valid and invalid argument. Plato had investigated truth, necessary connexion, definition, negation, and predication.

What had not yet appeared was a systematic formal theory tying such logical discoveries together.

Aristotle's decisive contribution is therefore best understood as **systematization**.



The achievement is enormous, but the Kneales also emphasize an important qualification.

The framework Aristotle systematizes is centered upon **general subject–predicate propositions**. That choice makes his theory extraordinarily powerful in one domain while also narrowing the kinds of logical form he develops.

Our task is therefore to understand both:

1. what Aristotle discovered; and
2. what the structure of his system caused him to overlook.

I. From Dialectic to Logical Theory

2 What Is the Organon?

The name *Organon* means roughly “instrument.”

It was not Aristotle’s title for a single book written according to one unified plan. After Aristotle’s death, a number of his works concerned with reasoning were grouped together, and the collection eventually came to be known as the *Organon*.

The modern word “logic” was not yet Aristotle’s name for the discipline. Nevertheless, the later conception of logic was deeply shaped by the works included in this collection.

The principal works relevant to our discussion are:

- the *Categories*;
- the *Topics*;
- the *De Sophisticis Elenchis*;
- the *De Interpretatione*;
- the *Prior Analytics*;
- the *Posterior Analytics*.

ARISTOTLE’S ORGANON

<i>Categories</i>	→	kinds / predication
<i>Topics</i>	→	dialectical argument
<i>Sophistical Refutations</i>	→	fallacious argument
<i>De Interpretatione</i>	→	statements / opposition
<i>Prior Analytics</i>	→	syllogistic form
<i>Posterior Analytics</i>	→	demonstration / science

The Organon is a collection, not one work written to a single plan.

The chronology of Aristotle’s works is difficult to determine. Many appear to have originated as lecture materials and may have been revised repeatedly.

For our purposes, however, a conceptual development is more important than exact chronology.

3 A Development Within the Organon

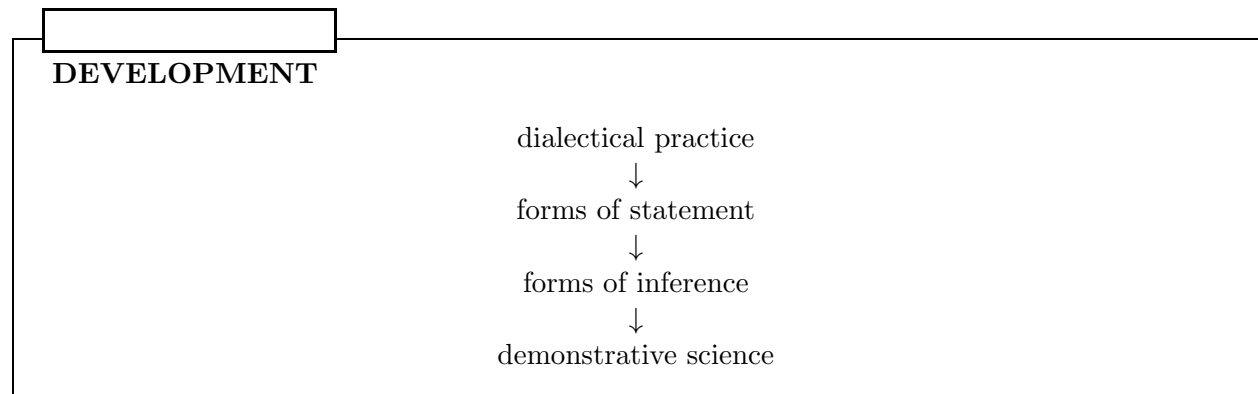
The works reveal a movement from practical concern with debate toward increasingly abstract reflection on logical form.

The *Topics* is still closely connected with dialectical contests.

The *De Interpretatione* isolates forms and oppositions of statements.

The *Prior Analytics* develops the systematic formal theory of syllogistic inference.

The *Posterior Analytics* considers the special conditions required for scientific demonstration.



This gives us the central historical movement of the chapter:

Aristotle moves from studying how people argue to studying the forms that make arguments necessary.

4 The Categories

The *Categories* is difficult to classify as a work of logic in the modern sense.

Much of it appears metaphysical. Nevertheless, its influence on later logic was enormous.

Aristotle presents ten highest categories:

1. substance,
2. quantity,
3. quality,
4. relation,
5. place,
6. time,
7. situation,
8. state,

9. action,
10. passion.

These are intended as extremely general kinds under which things signified by simple expressions can fall.

THE TEN CATEGORIES

Substance	Relation
Quantity	Place
Quality	Time
Situation	State
Action	Passion

Categories classify kinds of being / what terms signify.

We will not analyze each category separately. The most important for the history of Aristotle's logic is **substance**.

5 Primary and Secondary Substance

Aristotle distinguishes two kinds of substance.

A **primary substance** is an individual thing, such as an individual man or horse.

A **secondary substance** is the species or genus under which such an individual falls.

Thus we can represent the hierarchy:

Socrates → man → animal.

Socrates is an individual primary substance.

Man is a species.

Animal is a genus.

SUBSTANCE

Primary substance:	individual e.g. Socrates, this horse
Secondary substance:	species / genus e.g. man, animal

Socrates → man → animal

Primary substance is the ultimate subject.

The doctrine matters logically because Aristotle treats primary substances as the basic subjects upon which other things depend.

6 Words or Things?

The *Categories* contains a basic ambiguity.

Is Aristotle classifying linguistic expressions, or the things those expressions signify?

His terminology can suggest either answer.

The Kneales argue that Aristotle's fundamental interest is in **types of being**, but that he uses the behavior of linguistic expressions as evidence for distinctions among those types.

A difficulty is that Greek did not always provide the convenient distinctions that later logical writing developed.

We can easily distinguish:

man, "man", humanity.

These correspond roughly to a thing or kind, a word, and an abstract character.

Such distinctions were not always clearly marked in Aristotle's language.

WORD / THING / PROPERTY

"man" = word
man = kind
humanity = abstract character

language → clue to types of being

word ≠ thing ≠ property

The distinction will return repeatedly in the history of logic.

7 Three Logical Consequences of the Categories

The Kneales identify three especially important consequences of the theory of categories for later logic.

7.1 The Dominance of Subject–Predicate Form

If primary substance is the ultimate subject, then the basic form of truth naturally appears to be:

This thing is such-and-such.

Logic therefore becomes strongly centered upon subject–predicate propositions.

1. SUBJECT–PREDICATE DOMINANCE

Subject is Predicate

Basic picture:

individual subject + attribute

This concentration eventually makes it harder to give adequate treatment to relations and propositions involving several independent quantifiers.

7.2 The Singular–General Distinction Is Blurred

Consider:

Socrates is a man

and

Man is an animal.

The first has an individual subject.

The second has a general term as subject.

Yet both can be treated within Aristotle’s theory as subject–predicate propositions involving substance.

2. SINGULAR / GENERAL

Socrates is a man

Man is an animal

These are logically different kinds of statement.

7.3 An Early Theory of Type Distinctions

Different kinds of entities permit different kinds of predicates.

For example, Aristotle maintains that substances do not admit opposites in the way certain qualities do, nor do they admit “more and less” in the way some qualities do.

This suggests the beginnings of what later philosophers would call **type distinctions**: restrictions on what can significantly be said of what.

3. TYPE DISTINCTIONS

Different kinds of things permit different kinds of predication.

grammatical sentence $\nrightarrow$ logically appropriate predication

The overall balance is therefore important.

LOGICAL EFFECTS OF THE CATEGORIES

1. Subject–predicate form dominates
2. Singular / general distinction is blurred
3. Early type distinctions appear

Strength	Cost
organizes predication	narrows logical form

8 The Topics: Logic Inside Dialectic

The *Topics* is openly connected with public dialectical debate.

A disputation has two principal roles:

- the answerer maintains a thesis;
- the questioner attempts to undermine it.

The work offers strategies for conducting such disputes.

Its title comes from the notion of a *topos*, or “place.”

A *topos* is a recurrent argumentative strategy or move that can be used across many different subject matters.

TOPOS

a reusable argumentative move

Not:

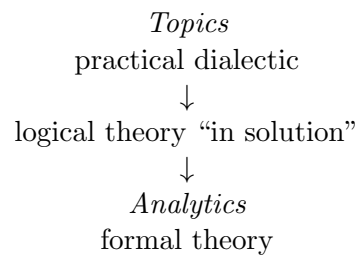
one particular argument

but:

a general strategy for argument

The Kneales characterize the *Topics* as containing logical theory “in solution.”

The more precise theories of the *De Interpretatione* and the *Prior Analytics* are what later crystallize out of it.



9 The Predicables

Much of the *Topics* concerns definition and classification.

Aristotle distinguishes four principal ways in which a predicate may stand to its subject.

9.1 Definition

A definition gives the essence of the subject: what it is to be that kind of thing.

9.2 Property

A property belongs uniquely and convertibly to a kind without constituting its essence.

9.3 Genus

A genus is the wider kind within which a species falls.

9.4 Accident

An accident may belong to instances of the kind but need not belong to them.

THE PREDICABLES

- Definition** → what the thing is
- Property** → unique, but not essence
- Genus** → wider kind
- Accident** → may belong; need not belong

Two underlying questions:

Necessary or non-necessary?

Convertible or non-convertible?

These distinctions contribute to Aristotle's later logical development.

10 Induction

The *Topics* also contains a distinction between syllogistic reasoning and induction.

Aristotle describes induction as a movement from particular or more specific cases toward the universal.

His examples need not be understood as mere enumeration of individual objects. They can involve comparison among specifically different cases.

INDUCTION

particular / specific cases
↓
universal

Example pattern:

skilled pilot → best pilot
skilled charioteer → best charioteer
⋮
∴ the skilled is best in its activity

Our main concern here is simply that Aristotle already distinguishes a movement toward universality from deductive reasoning from general premises.

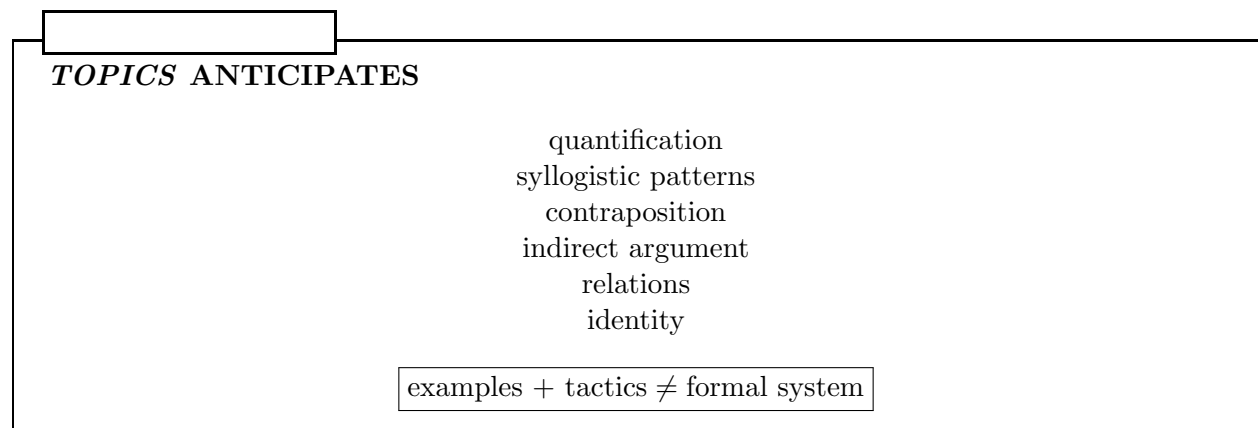
11 Logical Theory “in Solution”

The *Topics* contains many anticipations of Aristotle’s later formal theory.

Among them are:

- distinctions between universal and particular statements;
- patterns later incorporated into syllogistic;
- contraposition;
- indirect argument;
- principles concerning relations;
- principles of identity;
- arguments from “the more and the less.”

But these appear as examples and tactical rules rather than as parts of one formal calculus.



This is the point at which Aristotle’s logical thought is poised to become genuinely formal.

II. From Language to Proposition Form

12 Meaning: Words, Thoughts, and Things

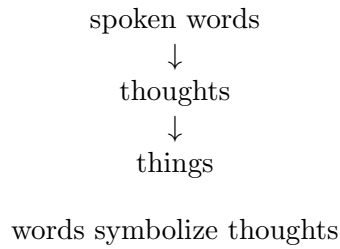
The opening chapters of the *De Interpretatione* continue questions already raised by Plato.

Aristotle distinguishes spoken words, mental experiences or thoughts, and the things to which those thoughts concern or correspond.

Spoken languages differ among peoples.

Aristotle nevertheless treats the underlying mental experiences and realities as common.

MEANING



A noun or verb by itself can be meaningful without being true or false.

For truth or falsity there must be some kind of composition.

“man” → meaningful
“runs” → meaningful
“Man runs” → capable of truth / falsity

Truth requires composition.

13 Declarative Discourse

Aristotle distinguishes declarative sentences from such utterances as prayers.

Declarative discourse is the kind he treats as capable of truth and falsity.

ARISTOTLE

declarative sentence → true or false
prayer / command → meaningful but not itself true or false

The Kneales argue that this restriction is too sharp for logic.

Commands, requests, questions, and similar utterances can express propositional material and stand in logical relations.

For example:

“Shut the door.”

and

“Do not shut the door.”

can conflict in a way that is logically intelligible.

KNEALES' QUALIFICATION

Different speech acts may still contain propositional content.

speech act $\neq$ propositional content

14 Aristotle on Truth

In the *Metaphysics*, Aristotle gives the famous correspondence-style account according to which truth consists, roughly, in saying of what is that it is and of what is not that it is not.

The basic logical point can be represented as follows.

TRUTH

Reality: P
Statement expresses P
 $\downarrow$
TRUE

"It is true that P " $\leftrightarrow P$

The apparently trivial character of this equivalence becomes important when Aristotle considers statements about the future.

15 Contradiction, Excluded Middle, and Bivalence

Three principles must be distinguished.

15.1 Law of Contradiction

The same proposition cannot be both true and false in the same respect.

In modern notation:

$$\neg(P \wedge \neg P).$$

15.2 Law of Excluded Middle

Either P or not- P :

$$P \vee \neg P.$$

15.3 Principle of Bivalence

Every proposition has one of two truth values:

true or false.

THREE PRINCIPLES

Contradiction: $\neg(P \wedge \neg P)$

Excluded Middle: $P \vee \neg P$

Bivalence: P is true or false

Do not simply identify Excluded Middle with Bivalence.

This distinction matters because of Aristotle's famous discussion of future contingents.

16 The Naval Battle

Consider the proposition:

P = "There will be a naval battle tomorrow."

The Law of Excluded Middle appears to give:

$$P \vee \neg P.$$

But suppose we also reason:

1. if P is already true now, the naval battle must occur;
2. if $\neg P$ is already true now, the naval battle must fail to occur;
3. therefore, whatever will happen tomorrow is already necessary;
4. therefore deliberation and action cannot alter the outcome.

This is the fatalistic argument that troubles Aristotle.

FUTURE CONTINGENT

P = “A naval battle will occur tomorrow.”

Today:

$$P \vee \neg P$$

Question:

Does one side already have to be true?

Aristotle rejects the fatalistic conclusion.

The Kneales’ diagnosis is that the crucial error is not the disjunction $P \vee \neg P$, but the movement from truth to necessity.

If it is true that a naval battle will occur, then a naval battle will occur.

But that does not mean that the occurrence is logically or metaphysically necessary merely because the corresponding proposition is true.

BAD MOVE

true $\implies$ necessary

TRUTH $\neq$ NECESSITY

“It is true that P ” $\leftrightarrow P$

Adding “it is true that” does not make P necessary.

The problem exposes a deeper issue: what exactly is the thing that is true or false?

17 Sentence, Statement, and Proposition

The Kneales distinguish several notions in order to diagnose the difficulty.

Four are especially important for us.

17.1 Token-Sentence

A token-sentence is one particular physical occurrence of a sentence: one utterance, inscription, or written occurrence.

17.2 Type-Sentence

A type-sentence is the repeatable linguistic pattern of which many tokens can be instances.

17.3 Statement

To make a statement is to use a declarative sentence in order to assert something.

17.4 Proposition

The proposition, or propositional content, is what is asserted.

DISTINGUISH

Token-sentence	particular utterance / inscription
Type-sentence	repeatable linguistic pattern
Statement	act of asserting
Proposition	what is asserted

Truth and falsity belong fundamentally to propositional content.

The same type-sentence can express different propositions on different occasions.

For example:

“I am sitting by the stove.”

can be uttered simultaneously by two people, one of whom is sitting by the stove and the other of whom is not.

The linguistic pattern is the same.

The proposition expressed is not.

same sentence-pattern $\nRightarrow$ same proposition

Truth is not fundamentally a property of mere words.

This clarifies why speaking of a proposition as becoming true or ceasing to be true can be misleading when what actually changes is the context in which a sentence is used.

18 Singular and General Statements

Aristotle distinguishes singular statements from general statements.

A singular statement has an individual subject:

Socrates is white.

A general statement concerns a kind:

Every man is white.

or

Some man is white.

STATEMENTS

Singular: Socrates is white.

General: Every man is white.
Some man is white.

Aristotle's mature formal logic concentrates on general statements.

The detailed syllogistic theory will be constructed primarily from general statements.

19 The Four Forms of General Statement

Two distinctions can be combined.

First:

affirmative vs. negative.

Second:

universal vs. particular.

Together these yield four forms.

The traditional labels *A, E, I, O* were introduced later in the Middle Ages. They are not Aristotle's own notation.

FOUR GENERAL FORMS

A Every *S* is *P*.

E No *S* is *P*.

I Some *S* is *P*.

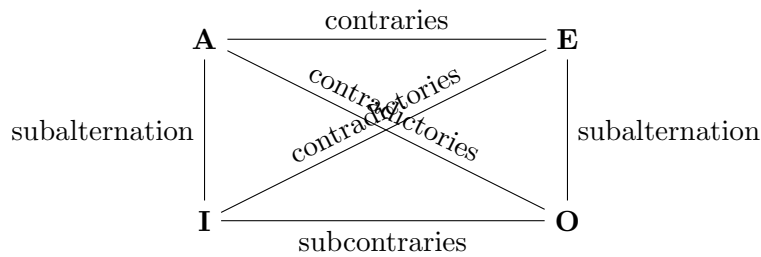
O Some *S* is not *P*.

	Affirmative	Negative
Universal	<i>A</i>	<i>E</i>
Particular	<i>I</i>	<i>O</i>

These four forms become the basic material of categorical syllogistic.

20 The Square of Opposition

The traditional square of opposition is not itself drawn by Aristotle, but it conveniently summarizes relations that he recognizes.



The contradictory pairs are:

$$A \leftrightarrow O$$

and

$$E \leftrightarrow I.$$

Contradictories cannot both be true and cannot both be false.

Contraries cannot both be true, though they may both be false.

Subcontraries may both be true, but within the traditional Aristotelian framework they are not both false.

Finally, Aristotle treats each universal statement as entailing its corresponding particular.

SQUARE OF OPPOSITION

$$\begin{array}{ccc} A & \longleftrightarrow & E \\ \uparrow & & \uparrow \\ I & \longleftrightarrow & O \end{array}$$

$A \leftrightarrow O$ contradictories

$E \leftrightarrow I$ contradictories

$A \leftrightarrow E$ contraries

$I \leftrightarrow O$ subcontraries

$A \Rightarrow I, \quad E \Rightarrow O$

These relations generate an important problem.

21 Existential Import

Why should

Every S is P

entail

Some S is P ?

That inference is valid only if something is in fact an S .

Aristotle's logic behaves as though the general terms under consideration have application.

EXISTENTIAL IMPORT

Aristotle's framework assumes that general terms have application.

Every S is P

$\therefore$ Some S is P

requires:

$\exists S$

The Kneales point out that modern logic need not make this assumption.

Consider:

“No mathematician has squared the circle.”

Rewriting it mechanically as

“No mathematician is a person who has squared the circle”

must not lead us to assume that there actually exists a person who has squared the circle.

MODERN CAUTION

No mathematician has squared the circle

does not imply:

$\exists x(x \text{ has squared the circle})$

Existence should be asserted explicitly when it matters.

The traditional square therefore works most naturally under Aristotle’s tacit assumption that its general terms have application.

22 Conversion

Before Aristotle develops syllogistic, he studies **conversion**.

Conversion interchanges the subject and predicate terms.

Certain forms convert simply.

Universal negatives:

No S is P

convert to

No P is S .

Particular affirmatives:

Some S is P

convert to

Some P is S .

Universal affirmatives convert only partially within Aristotle's system:

Every S is P

allows

Some P is S ,

provided the relevant existential assumptions hold.

Particular negatives do not simply convert.

CONVERSION

E : No S is $P \leftrightarrow$ No P is S

I : Some S is $P \leftrightarrow$ Some P is S

A : Every S is $P \Rightarrow$ Some P is S

O : no simple conversion

Conversion = interchange subject and predicate.

Conversion will become one of Aristotle's principal tools for showing that syllogisms in different figures are valid.

23 The Introduction of Variables

The *Prior Analytics* introduces one of Aristotle's most important technical innovations: the use of letters as **term variables**.

Earlier logical generality had often been expressed through examples.

For instance:

Every man is an animal.

Every Greek is a man.

∴ Every Greek is an animal.

But the logical subject matter is clearer if we abstract from “Greek,” “man,” and “animal.”

FROM EXAMPLE TO FORM

$$\begin{array}{l} \text{Every man is an animal.} \\ \text{Every Greek is a man.} \\ \hline \therefore \text{Every Greek is an animal.} \end{array}$$

becomes:

$$\begin{array}{l} \text{Every } M \text{ is } L. \\ \text{Every } S \text{ is } M. \\ \hline \therefore \text{Every } S \text{ is } L. \end{array}$$

VARIABLES expose FORM.

The Kneales describe this use of variables as an epoch-making advance in logical technique.

The letters do not stand for particular individuals. They mark places into which general terms can be substituted.

III. The Syllogistic System

24 What Is a Syllogism?

At the beginning of the *Prior Analytics*, Aristotle gives a very broad definition of syllogism.

A syllogism is discourse in which, certain propositions having been laid down, something different follows from them **of necessity**.

In the *Topics*, “syllogism” could therefore have a wide meaning.

But Aristotle’s detailed theory in the *Prior Analytics* becomes much narrower.

The characteristic categorical syllogism contains:

- two premisses;
- three general terms;
- a conclusion;
- a middle term connecting the other two terms.

SYLLOGISM

$$\begin{array}{c} \text{premiss} \\ \text{premiss} \\ \hline \text{conclusion} \end{array}$$

The conclusion follows OF NECESSITY.

Three terms:

L = major, S = minor, M = middle

M occurs in both premisses but not in the conclusion.

In our notation, L is the predicate term of the conclusion, S its subject term, and M the middle term.

This is a convenient later convention, not Aristotle's original lettering.

25 From Plato's Division to Aristotle's Syllogism

Aristotle criticizes Plato's method of division as a method of proof.

Suppose we divide animals into mortal and immortal animals, and then place humans under the mortal branch.

That procedure may classify humans as mortal animals, but if the classification is simply assumed in the division, it does not prove that humans are mortal.

Division therefore risks begging the very point that requires proof.

PLATO VS. ARISTOTLE

$$\begin{array}{ll} \text{Division} & \rightarrow \text{classification} \\ \text{Syllogism} & \rightarrow \text{proof} \end{array}$$

L applies to M

M applies to S

$\therefore L$ applies to S

Although Aristotle rejects division as proof, the Kneales suggest that the classificatory chain inherited from Plato helped shape his conception of the syllogism.

26 The Three Figures

Aristotle distinguishes three figures according to the position of the middle term.

Using our standardized notation:

26.1 First Figure

$$\begin{array}{c} M - L \\ S - M \\ \hline S - L \end{array}$$

The middle term is subject in one premiss and predicate in the other.

26.2 Second Figure

$$\begin{array}{c} L - M \\ S - M \\ \hline S - L \end{array}$$

The middle term is predicate in both premisses.

26.3 Third Figure

$$\begin{array}{c} M - L \\ M - S \\ \hline S - L \end{array}$$

The middle term is subject in both premisses.

THE THREE FIGURES

1	2	3
$M - L$	$L - M$	$M - L$
$S - M$	$S - M$	$M - S$
$S - L$	$S - L$	$S - L$

The position of the MIDDLE TERM determines the figure.

Aristotle recognizes fourteen valid moods distributed across these figures.

27 The Four First-Figure Moods

The medieval names for the moods were introduced later.

Nevertheless, they remain convenient labels.

The four fundamental first-figure moods are:

27.1 Barbara

$$\begin{array}{l} \text{Every } M \text{ is } L. \\ \text{Every } S \text{ is } M. \\ \hline \therefore \text{Every } S \text{ is } L. \end{array}$$

27.2 Celarent

$$\begin{array}{l} \text{No } M \text{ is } L. \\ \text{Every } S \text{ is } M. \\ \hline \therefore \text{No } S \text{ is } L. \end{array}$$

	FIRST FIGURE I
--	-----------------------

BARBARA

$$\begin{array}{l} \text{Every } M \text{ is } L. \\ \text{Every } S \text{ is } M. \\ \hline \therefore \text{Every } S \text{ is } L. \end{array}$$

CELARENT

$$\begin{array}{l} \text{No } M \text{ is } L. \\ \text{Every } S \text{ is } M. \\ \hline \therefore \text{No } S \text{ is } L. \end{array}$$

Copy, then erase.

27.3 Darii

$$\begin{array}{l} \text{Every } M \text{ is } L. \\ \text{Some } S \text{ is } M. \\ \hline \therefore \text{Some } S \text{ is } L. \end{array}$$

27.4 Ferio

$$\begin{array}{l} \text{No } M \text{ is } L. \\ \text{Some } S \text{ is } M. \\ \hline \therefore \text{Some } S \text{ is not } L. \end{array}$$

FIRST FIGURE II

DARII

$$\begin{array}{l} \text{Every } M \text{ is } L. \\ \text{Some } S \text{ is } M. \\ \hline \therefore \text{Some } S \text{ is } L. \end{array}$$

FERIO

$$\begin{array}{l} \text{No } M \text{ is } L. \\ \text{Some } S \text{ is } M. \\ \hline \therefore \text{Some } S \text{ is not } L. \end{array}$$

Aristotle recognizes fourteen valid moods in all.

28 Reference: Aristotle's Fourteen Valid Moods

The following table is included for reference. Students need not memorize the medieval names at this stage.

Figure	Mood	Form
1	Barbara	Every M is L ; Every S is M ; therefore Every S is L .
1	Celarent	No M is L ; Every S is M ; therefore No S is L .
1	Darii	Every M is L ; Some S is M ; therefore Some S is L .
1	Ferio	No M is L ; Some S is M ; therefore Some S is not L .
2	Cesare	No L is M ; Every S is M ; therefore No S is L .
2	Camestres	Every L is M ; No S is M ; therefore No S is L .
2	Festino	No L is M ; Some S is M ; therefore Some S is not L .
2	Baroco	Every L is M ; Some S is not M ; therefore Some S is not L .
3	Darapti	Every M is L ; Every M is S ; therefore Some S is L .
3	Felapton	No M is L ; Every M is S ; therefore Some S is not L .
3	Disamis	Some M is L ; Every M is S ; therefore Some S is L .
3	Datisi	Every M is L ; Some M is S ; therefore Some S is L .
3	Bocardo	Some M is not L ; Every M is S ; therefore Some S is not L .
3	Ferison	No M is L ; Some M is S ; therefore Some S is not L .

Darapti and Felapton depend upon Aristotle's assumption of existential import. Without an assumption that there actually are M 's, their particular conclusions do not follow from their two universal premisses.

29 Worked Example: Barbara

Consider:

Every mammal is an animal.

Every dog is a mammal.

Therefore:

Every dog is an animal.

BARBARA: CONCRETE FORM

Every mammal is an animal.

Every dog is a mammal.

∴ Every dog is an animal.

Now replace the particular content with variables.

BARBARA: ABSTRACT FORM

Every M is L .

Every S is M .

∴ Every S is L .

CONTENT changes; **FORM** remains.

The variables reveal precisely what makes the inference independent of its subject matter.

30 Why the First Figure Is “Perfect”

Aristotle calls first-figure syllogisms **perfect**.

The likely reason is that their necessity becomes evident from the arrangement of terms without requiring supplementary transformations.

Barbara can be visualized pedagogically as:

$$S \subseteq M$$

and

$$M \subseteq L,$$

hence:

$$S \subseteq L.$$

This class notation is a modern explanatory device, not Aristotle's symbolism.

WHY FIGURE 1 IS "PERFECT"

$$S \subseteq M$$

$$M \subseteq L$$

$$\therefore S \subseteq L$$

The transitivity of the term connexion is visible.

The other figures require additional work to show that their conclusions follow.

31 The Special Character of the Three Figures

The Kneales summarize the figures in useful conceptual terms.

In the **first figure**, the reasoner establishes a conclusion by showing that a sufficient condition is fulfilled.

In the **second figure**, the conclusion must be negative. The argument shows that a necessary condition is not fulfilled.

In the **third figure**, the conclusion must be particular. The argument proceeds through instances.

THREE FIGURES: LOGICAL CHARACTER

Figure 1: sufficient condition fulfilled

Figure 2: necessary condition fails

Figure 3: instances establish a particular

This interpretation also helps explain why existential assumptions are particularly important in the third figure.

32 How Aristotle Rejects Invalid Forms

Aristotle does not merely list valid moods.

He also considers pairs of premiss forms that fail to determine any necessary conclusion.

One important technique is what the Kneales call **rejection by contrasted instances**.

Suppose the premisses have the form:

Every M is L

and

No S is M .

Does anything follow about the relation between S and L ?

Consider first:

$M = \text{man}, \quad L = \text{animal}, \quad S = \text{horse}.$

The premisses are true:

Every man is an animal

and

No horse is a man.

Yet:

Every horse is an animal.

Now change only S :

$S = \text{stone}.$

Again the premisses are true:

Every man is an animal

and

No stone is a man.

But:

No stone is an animal.

The same premiss form is therefore compatible with opposite relations between S and L .

CONTRASTED INSTANCES

Premiss form:

Every M is L

No S is M

Case 1:

$M = \text{man}, \quad L = \text{animal}, \quad S = \text{horse}$

Every horse is an animal

Case 2:

$M = \text{man}, \quad L = \text{animal}, \quad S = \text{stone}$

No stone is an animal

Therefore the original premiss structure determines neither an affirmative nor a negative conclusion.

NO NECESSARY CONCLUSION

Counterexamples can refute a proposed argument form.

This is an important early use of counterexample-style reasoning in formal logic.

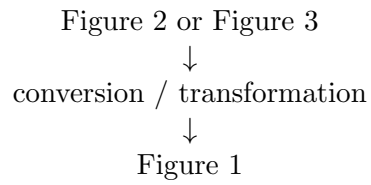
33 Reduction

Aristotle's fourteen valid moods are not merely a catalogue.

He organizes the system through the doctrine of **reduction**.

Syllogisms in the second and third figures are validated by transforming them into first-figure syllogisms.

REDUCTION



“Imperfect” moods are validated through “perfect” moods.

For most moods, conversion is enough.

Two moods, however, require an indirect procedure:

- Baroco;
- Bocardo.

For these Aristotle uses what later came to be called *reduction per impossibile*.

34 Worked Example: Indirect Reduction

Consider the Baroco form:

Every L is M .

Some S is not M .

Therefore:

Some S is not L .

INDIRECT REDUCTION: PACKET 1

Given:

Every L is M

Some S is not M

Want:

Some S is not L

Assume the contradictory:

Every S is L

Copy, then erase.

Now combine:

Every L is M

with

Every S is L .

Barbara gives:

Every S is M .

But the original premiss says:

Some S is not M .

Contradiction.

Therefore the denial of the original conclusion cannot stand.

INDIRECT REDUCTION: PACKET 2

Every L is M

Every S is L

$\therefore$ Every S is M

But:

Some S is not M

Therefore:

$\perp$

and hence:

Some S is not L

REDUCTION PER IMPOSSIBILE

The procedure connects Aristotle's formal system directly with the older Greek tradition of reductio.

35 Ecthesis

Aristotle also sometimes uses a method called **ecthesis**, literally a "setting out."

The basic procedure is to introduce an appropriate instance in order to reason about a general relation.

ECTHESIS

general claim
 $\downarrow$
set out an appropriate instance
 $\downarrow$
reason from that instance

Ancient and modern scholars have disputed whether Aristotle's introduced item should always be understood as an individual or sometimes as a subclass.

For our purposes, the important point is simply that ecthesis is another technique by which Aristotle validates logical transformations.

36 Aristotle's System Is Not a Modern Axiomatic System

Modern textbooks often present Barbara like this:

$$\frac{\text{Every } M \text{ is } L. \\ \text{Every } S \text{ is } M.}{\therefore \text{Every } S \text{ is } L.}$$

Aristotle's own characteristic formulation is instead a conditional principle:

If every M is L and every S is M ,

then necessarily every S is L .

Later antiquity increasingly presented the theory through inference schemata.

The difference should not be exaggerated, since Aristotle uses his conditional principles much as rules of inference.

But historically we should not simply reconstruct his theory as though it were already a modern formal calculus.

ARISTOTLE'S PRESENTATION

If P_1 and P_2 , then necessarily C .

Later presentation:

$$\frac{P_1 \\ P_2}{C}$$

Systematic? YES	Modern axiomatic calculus? NO
-----------------	-------------------------------

Later logicians, most notably Jan Lukasiewicz, successfully reconstructed Aristotelian syllogistic as an axiomatic deductive system.

That reconstruction is logically interesting but should not be confused with Aristotle's own conception of his work.

IV. Beyond Pure Assertoric Syllogistic

37 Modal Logic

Aristotle does not limit himself to ordinary, or **assertoric**, propositions.

He also attempts a theory of modal propositions and modal syllogisms.

A modal proposition contains such notions as necessity or possibility.

The three broad kinds relevant to Aristotle's treatment are:

- assertoric;
- apodeictic;
- problematic.

MODAL STATEMENTS

Assertoric: P

Apodeictic: necessarily P

Problematic: possibly / contingently P

Modal logic asks how modality changes inference.

The Kneales judge Aristotle's modal syllogistic to be much less successful than his assertoric syllogistic.

The theory is complicated, lacks sufficiently clear symbolism, and is developed without a prior systematic logic of propositions.

Nevertheless, Aristotle's decision to treat modality logically is itself a major achievement.

38 Necessity, Possibility, and Contingency

The Kneales distinguish two notions that Aristotle expresses with closely related terminology.

In the broad sense:

possible P

means:

not impossible P .

But Aristotle's stricter problematic notion is better translated by the Kneales as **contingent**.

A proposition is contingent when it is possible but not necessary.

Thus:

$$\text{Contingent } P = \text{Possible } P \wedge \neg \text{Necessary } P.$$

MODAL RELATIONS

$$\text{Necessary } P \Rightarrow \text{Possible } P$$

$$\text{Impossible } P = \text{Necessary } \neg P$$

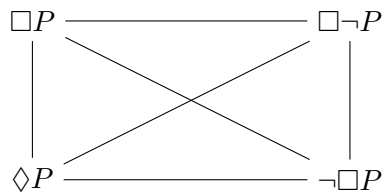
$$\text{Possible } P = \neg \text{Necessary } \neg P$$

$$\text{Contingent } P = \text{Possible } P \wedge \neg \text{Necessary } P$$

One difficulty in Aristotle's modal syllogistic is precisely that contingency is already logically complex.

39 The Modal Square

The modal notions can be arranged by analogy with the ordinary square of opposition.



Here:

$$\Box P$$

means that P is necessary,

and:

$$\Diamond P$$

means that P is possible.

Contingency is represented by the conjunction:

$$\Diamond P \wedge \neg \Box P.$$

MODAL SQUARE

Necessary P	Impossible P
Possible P	Not-necessary P

$$\text{Contingent } P = \text{Possible } P + \text{Not-necessary } P$$

40 Separately Possible Does Not Mean Jointly Possible

One source of difficulty in modal reasoning is the assumption that two things that are separately possible must also be possible together.

But:

$$\Diamond P$$

and

$$\Diamond Q$$

do not in general entail:

$$\Diamond(P \wedge Q).$$

MODAL CAUTION

$$\Diamond P$$

$$\Diamond Q$$

does **not** guarantee:

$$\Diamond(P \wedge Q)$$

SEPARATELY POSSIBLE $\neq$ JOINTLY POSSIBLE

This kind of problem helps explain why problematic syllogistic cannot simply be obtained by inserting modal words into ordinary syllogisms.

41 The Scope of Modality

Another major difficulty concerns where the modal qualification belongs.

Compare:

$$\Box(\text{Every } B \text{ is } A)$$

with:

Every B is necessarily- A .

These are not automatically equivalent.

The modal qualification should not simply be treated as though it were an ordinary component of the predicate.

SCOPE MATTERS

Correct logical structure:

$$\Box(\text{Every } B \text{ is } A)$$

Do not automatically rewrite as:

Every B is $\Box A$

The modality qualifies the proposition as a whole.

The Kneales use this distinction to diagnose some of Aristotle's unsuccessful modal syllogisms.

42 Why Modality Belongs in Logic

Aristotle's technical modal theory is imperfect.

But the Kneales argue that Aristotle was entirely right to regard modality as part of logic.

There are two reasons.

First, distinctions between:

true, necessary, false, impossible

apply across subject matters.

Second, the very notion of validity is naturally expressed in modal language.

An argument is valid when it is **impossible** for its premisses to be true while its conclusion is false.

VALIDITY

A valid argument makes this impossible:

premisses true $\wedge$ conclusion false

Symbolically:

$$\neg \Diamond (\text{Premisses true} \wedge \text{Conclusion false})$$

Logic itself uses modality.

Aristotle therefore deserves credit for initiating logical inquiry into modality even where the resulting details are defective.

43 Non-Syllogistic Logic in the Analytics

The *Prior Analytics* contains logical principles that do not belong to categorical syllogistic.

The Kneales identify two especially important principles that can be expressed in modern terminology as **contraposition** and **transitivity of entailment**.

43.1 Contraposition

If P entails Q , then the falsity of Q entails the falsity of P .

$$P \Rightarrow Q$$

therefore:

$$\neg Q \Rightarrow \neg P.$$

CONTRAPOSITION

$$P \Rightarrow Q$$

$$\therefore \neg Q \Rightarrow \neg P$$

43.2 Transitivity

If P entails Q , and Q entails R , then P entails R .

$$P \Rightarrow Q$$

$$Q \Rightarrow R$$

therefore:

$$P \Rightarrow R.$$

TRANSITIVITY OF ENTAILMENT

$$P \Rightarrow Q$$

$$Q \Rightarrow R$$

$$\therefore P \Rightarrow R$$

These principles concern propositions, not merely relations among terms.

These passages are important because they show that Aristotle's logical insights extend beyond the formal boundaries of his syllogistic system.

44 “Either Way, We Must Philosophize”

The Kneales also preserve an argument from Aristotle's early *Protrepticus*.

Its idea is:

- either we ought to philosophize or we ought not;
- if we ought, then we ought to philosophize;
- if we ought not, philosophical reasoning is still needed to justify that claim;
- therefore, in either case, we must philosophize.

Abstractly, the form is:

$$P \vee \neg P$$

$$P \Rightarrow Q$$

$$\neg P \Rightarrow Q$$

therefore:

$$Q.$$

EITHER WAY

$$P \vee \neg P$$

$$P \Rightarrow Q$$

$$\neg P \Rightarrow Q$$

$$\therefore Q$$

Application:

Either way, we must philosophize.

This kind of reasoning does not fit naturally into categorical syllogistic.

It points toward a logic whose units are whole propositions.

45 Reasoning From Hypothesis

Aristotle recognizes arguments “from hypothesis,” but he does not build a systematic logic of conditionals comparable to his theory of syllogisms.

Two forms that later become fundamental are:

$$\text{If } P \text{ then } Q; \quad P; \quad \therefore Q$$

and:

$$\text{If } P \text{ then } Q; \quad \neg Q; \quad \therefore \neg P.$$

CONDITIONAL ARGUMENT FORMS

$$\frac{P \rightarrow Q \quad P}{Q}$$

and

$$\frac{P \rightarrow Q \quad \neg Q}{\neg P}$$

The second is the form underlying many reductio arguments.

The Kneales' criticism is not that Aristotle never uses such reasoning.

He plainly does.

The limitation is that he does not isolate and systematize the logical distinction between these forms in the way later propositional logicians will.

46 Theophrastus

Aristotle's logical project continued immediately within his own school.

Theophrastus succeeded Aristotle as head of the Lyceum.

His logical works no longer survive intact, so our knowledge depends heavily on later reports.

Nevertheless, he appears to have extended Aristotle's work in several directions.

THEOPHRASTUS

Extends Aristotle in:

1. syllogistic
2. modal logic
3. hypothetical logic

Aristotle $\longrightarrow$ Theophrastus $\longrightarrow$ later tradition

47 Indirect Moods and the Later Fourth Figure

Theophrastus is reported to have recognized additional indirect forms associated with Aristotle's first figure.

Later tradition reorganized such forms as a **fourth figure**.

Its characteristic pattern may be represented:

$$\frac{L - M}{\frac{M - S}{S - L}}$$

LATER “FOURTH FIGURE”

$$\frac{L - M}{\frac{M - S}{S - L}}$$

Aristotle : 3 figures
later tradition : 4 figures

Whether this is a separate figure depends upon how “figure” is defined.

The point is historically interesting but less important than Theophrastus’ contribution to hypothetical reasoning.

48 Theophrastus and Modal Logic

Aristotle’s immediate successors already recognized difficulties in his modal syllogistic.

Theophrastus appears to have revised parts of the system.

Because the ancient evidence is fragmentary, we should avoid pretending that we possess a complete reconstruction.

The important historical point is the speed with which formalization generated criticism and revision.

MODAL SYLLOGISTIC

Aristotle
↓
technical difficulties
↓
Theophrastus revises

Formalization exposed problems that informal reasoning had hidden.

49 Hypothetical Reasoning

In the ancient Peripatetic tradition, the word “hypothetical” has a broader range than our modern word “conditional.”

It can include reasoning involving complex premisses such as:

$$P \rightarrow Q,$$

$$P \vee Q,$$

and

$$\neg(P \wedge Q).$$

“HYPOTHETICAL”

In this tradition:

hypothetical argument $\approx$ argument with a complex premiss

Examples:

$$P \rightarrow Q$$

$$P \vee Q$$

$$\neg(P \wedge Q)$$

This points beyond categorical syllogistic.

The evidence does not allow us to reconstruct a complete Aristotelian hypothetical logic, but the material becomes increasingly important in Aristotle’s school.

50 A Conditional Argument

A reported example has the form:

If the soul always moves, the soul is immortal.

The soul always moves.

Therefore:

The soul is immortal.

The logical pattern is:

$$P \rightarrow Q$$

$$P$$

$$\therefore Q.$$

CONDITIONAL INFERENCE

$$\frac{P \rightarrow Q \quad P}{Q}$$

conditional premiss + additional premiss $\rightarrow$ conclusion

We should not infer from this that Aristotle or Theophrastus had already produced the mature Stoic propositional logic.

It does show that the Peripatetic tradition was developing logical patterns not reducible to ordinary term syllogisms.

51 Theophrastus and Inference Schemata

Later testimony credits Theophrastus with systematic treatment of what were called totally hypothetical syllogisms.

One form is:

$$P \rightarrow Q$$

$$Q \rightarrow R$$

therefore:

$$P \rightarrow R.$$

THEOPHRASTUS

$$P \rightarrow Q$$

$$Q \rightarrow R$$

$$\therefore P \rightarrow R$$

transitivity of implication

The Kneales emphasize another historically interesting point.

Aristotle usually states the logical principle corresponding to a mood as a conditional sentence:

If the premisses hold, the conclusion necessarily follows.

Theophrastus appears to have presented hypothetical reasoning through something much closer to an explicit **inference schema**.

FROM TERM PATTERNS TO PROPOSITION PATTERNS

Aristotelian emphasis:

$$S, M, L$$

Theophrastean hypothetical form:

$$P, Q, R$$

Whole propositions become the units of the pattern.

This development points directly toward the Megarian and Stoic tradition.

52 Prosleptic Reasoning

Theophrastus is also associated with **proslaptic syllogisms**.

The important conceptual feature is that such reasoning can generalize over kinds or predicates themselves rather than merely over individual things.

A source example can be represented approximately as follows:

- substance is predicable of whatever is universally predicable of man;

- animal is universally predicable of man;
- therefore substance is predicable of animal.

The abstract structure is more important than the terminology.

PROSLEPTIC REASONING

Ordinary syllogistic:

generalizes over individuals

Prosleptic pattern:

generalizes over kinds / predicates

Abstractly:

$$\frac{\text{Whatever satisfies } C \text{ has } A}{B \text{ satisfies } C} \quad \therefore B \text{ has } A$$

The Kneales compare aspects of this with later ideas of substitution and detachment.

53 What Aristotle Achieved

Aristotle's achievement can now be stated more precisely.

He provides:

- an explicit classification of general propositions;
- a systematic account of opposition;
- the use of logical variables;
- a formal theory of categorical syllogisms;
- a classification into figures and moods;
- techniques for rejecting invalid forms;
- conversion;
- reduction of imperfect to perfect syllogisms;
- indirect reduction;
- an attempt at modal logic;
- principles of entailment beyond syllogistic.

ARISTOTLE'S ACHIEVEMENT

- ✓ quantified proposition forms
- ✓ variables
- ✓ systematic syllogistic
- ✓ valid / invalid forms
- ✓ reduction
- ✓ modal inquiry
- ✓ general entailment principles

These discoveries justify Aristotle's extraordinary position in the history of logic.

54 The Limits of Aristotle's Framework

The same chapter also reveals important limitations.

Aristotle's conception of the proposition is dominated by the subject–predicate structure.

His system does not sharply enough distinguish singular from general propositions.

It tacitly assumes application for general terms in places where modern logic makes existence explicit.

Relations receive only scattered treatment.

Complex propositions are not systematically developed into a logic of propositions.

And the modal syllogistic contains serious technical problems.

LIMITS

- × subject–predicate dominance
- × singular/general distinction blurred
- × existential assumptions
- × relations underdeveloped
- × no complete propositional logic
- × problematic modal system

Yet this is the first systematic formal logic.

The limitations do not diminish the achievement.

They explain the direction in which subsequent logical development had to proceed.

55 Two Directions After Aristotle

By the end of the chapter, two lines of development have become visible.

The first is Aristotle's great achievement:

categorical syllogistic.

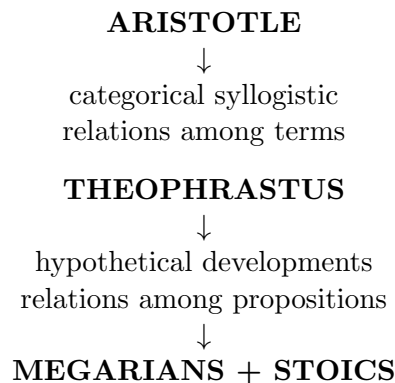
Its primary objects are relations among general terms.

The second emerges through material Aristotle does not fully systematize and through the work of Theophrastus:

hypothetical and propositional reasoning.

Its primary objects increasingly become relations among whole propositions.

TWO DIRECTIONS



This gives us the transition to the next chapter.

56 Lecture Summary

Aristotle inherited a Greek tradition in which thinkers already reasoned about demonstration, dialectic, fallacy, truth, definition, and necessary connexion.

His achievement was to transform this material into a systematic investigation of logical form.

The *Categories* supplied a metaphysical and linguistic framework centered on predication.

The *Topics* contained logical theory still embedded in dialectical practice.

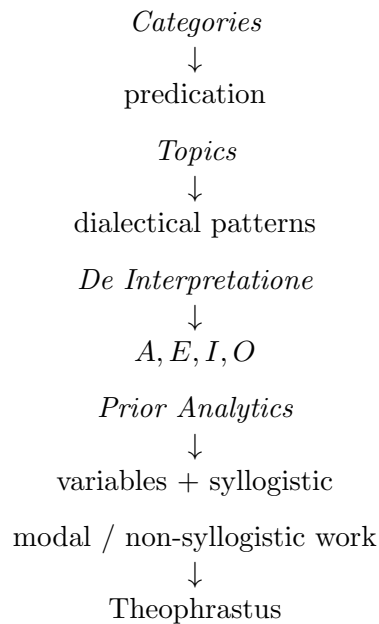
The *De Interpretatione* isolated forms of statement and their oppositions.

The *Prior Analytics* introduced variables and created the first systematic theory of syllogistic inference.

Aristotle then attempted to extend formal logic to modality and also stated principles of entailment that reached beyond the limits of his categorical theory.

Theophrastus continued the project and pushed the Peripatetic tradition toward increasingly explicit hypothetical reasoning.

CHAPTER II SUMMARY



FINAL TAKEAWAY

Aristotle did not invent reasoning.
He invented a systematic formal theory
for studying important forms of reasoning.

Next: The Megarians and the Stoics